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Multi-label chest X-ray image classification based on long-range dependencies capture and label relationships learning

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ABSTRACT

Diagnosing chest diseases from X-ray images using convolutional neural networks (CNNs) is an active area of research. However, existing methods mostly focus on extracting feature information from local regions for prediction, while ignoring the larger-scale image contextual information. Moreover, anatomical segmentation knowledge and co-occurrence relationships among labels, which are important for classification, are not fully utilized. To address the above problems, we proposed a method to capture long-range dependent information in chest X-ray images using a CNN with large kernel convolution. Furthermore, it captures the detailed features of the interest region through anatomical segmentation and builds the potential relationships of different diseases using a graph convolutional network (GCN). Firstly, we pre-trained UNet from a dataset with organ-level annotations for segmenting anatomical regions of interest in the images. Secondly, we build a four-stage backbone network using the large kernel attention (LKA) mechanism and superimpose anatomically segmented regions on the feature maps of each stage to obtain different scales of feature maps for the regions of interest. Thirdly, we utilized a GCN to obtain a co-occurrence matrix representing the potential relationships between all disease labels in the training dataset. Finally, we get the disease diagnosis by combining the label co-occurrence matrix and the visual feature maps. We experimentally show that our proposed method achieves excellent AUC scores of 91.5%, 84.5%, and 82.5% on three publicly available CXR datasets–NIH, Stanford CheXpert, and MIMIC-CXR-JPG, respectively.

1. Introduction

Over the past decades, humans have made tremendous progress in Deep Learning techniques for medical image analysis driven by large medical datasets. Chest X-ray (CXR) imaging is a rapid, painless disease screening technique that provides rich visual diagnostic information and is one of the most commonly used imaging modalities for the diagnosis of critical diseases of the lungs and heart. Despite the large amount of chest imaging data generated each year, the lack of experts to analyze X-ray images [1], long screening times, and visual similarity of image disease clues [2], make it difficult to accurately and efficiently diagnose disease. Therefore, there is an urgent need to develop computer-aided and interpretable decision support systems to help medical personnel reliably identify pathological CXR images in clinical routine [3].

In the field of medical signal processing, more and more methods and techniques are being applied to extract and analyze information. Time-frequency analysis techniques [4] (e.g., wavelet transform and short-time Fourier transform) are widely used since they can provide information in both time and frequency. For example, time–frequency analysis in ECG and EEG analysis can effectively detect abnormalities such as cardiac arrhythmias and seizures [5,6], while in X-ray images it can help identify morphological changes in tumors or other lesions. The excellent time–frequency characteristics of wavelet transform determine that it can be used to improve image quality and remove noise [7–9]. Potential biological morphological structural features and pathological changes can be extracted by frequency domain analysis of medical images [10]. Although some studies based on time–frequency analysis feature extraction and bi-directional long-short-term memory (BiLSTM) models have demonstrated the potential of this technique for medical signal applications [11–13], conventional time–frequency analysis may suffer from challenges such as pattern mixing and noise sensitivity when confronted with complex patterns. In X-ray image processing, techniques such as deep learning may be more effective in improving classification and recognition performance due to the presence of subtle structural changes in pathologies.

Deep Learning based methods have already shown impressive

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performance in the field of medical image classification. However, most application scenarios usually involve only one specific type of disease or lesion, such as the automatic classification of tuberculosis [14–16], detection of COVID-19 [17–19], and segmentation of lung nodules [20,21]. However, multiple diseases may co-exist in a CXR image, and it becomes more complicated to accurately identify these diseases. Although physicians can diagnose diverse chest diseases with rich clinical knowledge, the complex correlations between these diseases are difficult to express clearly. Therefore, exploring how to effectively capture the correlations between disease labels is important for CXR image disease diagnosis.

Most of the present methods [22–27] used global learning strategies based on encoder-decoders and rely on local attention mechanisms to achieve feature extraction from images. Unfortunately, this often results in ignoring large-scale image context information, which is often critical for accurate classification. Dora et al. [28] proposed a multiple kernel based convolutional neural network (MK-CNN) for pathological brain classification. The method uses convolutional kernels of different sizes to extract multi-scale features, which can capture local detail information and retain global spatial information. Compared with the traditional transfer learning method, this method can better handle the case of small dataset and avoid the negative transfer and overfitting problems. Hu et al. [29] proposed the multi-kernel depthwise convolution (MD-Conv) module to learn multi-scale features by combining deep convolutional kernels of different sizes and expanding the receptive field by using larger convolutional kernels, which proved it suitable for diagnosing chest abnormalities of different sizes. These methods for achieving large-scale context fusion based on larger convolutional kernels can expand the receptive field of the model by integrating local and global features of the image. Thus, they can comprehensively evaluate pathological features in images.

In multi-label classification tasks, there are often complex correlations between labels, and traditional CNNs have limited performance in dealing with these correlations. To address this problem, several studies [30–35] utilized GCNs to capture the dependencies between labels relationships, thus enhancing the classification performance. GCNs can propagate information over graph structures and thus perform well in multi-label classification tasks, especially in capturing complex relationships between labels. Krishna et al. [36] proposed a spectral graph wavelet transform (SGWT)-based method for feature extraction and representation by considering using graph structures to represent interdependencies between different EEG signal samples. The application of SGWT has shown that the graph structure is very effective in finding multi-scale information at the local level as well as at the global level. In addition, some studies [27,37] incorporate medical a priori knowledge, such as prioritizing pre-identified features of specific anatomical regions. This draws on the diagnostic process of radiologists, who typically locate salient anatomical structures before determining the presence of abnormalities in these regions. Therefore, integrating anatomical knowledge into deep learning models can significantly improve automatic disease classification. Overall, the task of disease classification for CXR images still faces many challenges, such as the multi-label correlation problem, the effective fusion of local and global information, and the combination of medical prior knowledge. To address these problems, it is important to explore strategies that combine GCNs with traditional convolutional networks.

In this paper, our approach fuses large-scale contextual feature information in images and a priori knowledge of medical anatomy to obtain visual features that are more useful for chest disease diagnosis. Meanwhile, we utilize graph convolutional networks to solve the label correlation problem in multi-label classification tasks. The contributions of this paper are summarized as follows:

- We build a backbone network by the Large Kernel Attention (LKA) mechanism, which captures long-range dependencies feature information in medical images, allowing the network to comprehensively understand the image context.
- We performed feature masking of anatomical regions using a pre-trained UNet to focus on feature information in the lung and heart regions. We model the co-occurrence relationships between various diseases by GCN. Disease diagnosis results are obtained by integrating the visual features of the image modality and the labeled potential relationships of the text modality.
- We applied the proposed method to three popular chest disease classification datasets, NIH, Stanford CheXpert, and MIMIC-CXR-JPG, and achieved good results. These experiments demonstrate the robustness and generalization of our method, which helps to assist physicians in the diagnosis of chest diseases.

2. Related work

2.1. Convolutional neural networks

The existing studies in the fields of chest disease analysis generally utilize various architectures of classic CNNs such as VGGNet [38], ResNet [39], DenseNet [40], etc., to diagnose multiple diseases present on CXR datasets. CheXNet [22] suggested using the DenseNet121 architecture, which was the most advanced model for classification at that time. CheXNeXt [23] is an improved version of CheXNet, which exhibited the same level of accuracy as radiologists in a total of 10 lesions. Kumar et al. [24] proposed a cascaded deep neural network with DensNet161 as the baseline neural network architecture on ChestX-ray14 to improve the performance of multi-label classification tasks. Kumar et al. [41] proposed an algorithm for liver tumor detection combining fuzzy logic and neural networks. Fuzzy logic is used to process the uncertainty information extracted from medical images, while neural networks and windowing techniques automatically segment and filter to detect tumors. Jyoti et al. [42] decomposed CXR images using tunable Q-wavelet transform (TQWT) to improve the peak signal-to-noise ratio and structural similarity index and processed the decomposed images by memristive crossbar array (MCA) model for efficient storage, and subsequently utilized ResNet50and AlexNet to classify the processed images. Nayak et al. [43] evaluated the effectiveness of eight pre-trained Convolutional Neural Network (CNN) models for the classification of COVID-19 from normal cases. They found that the ResNet-34 model was the best model for COVID-19 detection by analyzing and comparing the hyperparameter tuning. Rubin et al. [25] introduced a novel dual convolutional network, DualNet, jointly trained on frontal and lateral CXR images from the MIMIC-CXR-JPG dataset. DualNet provides improved performance when classifying findings in CXR images compared to individual baseline (i.e. front or side) classifiers. Lanting Li et al. [35] proposed an end-to-end multi-label learning framework GL-MLL, for jointly modeling global and local label relevance, considering unbalanced class distributions, and focusing on capturing label-specific features in image-level representations. Pham et al. [44] introduced a multi-label classification framework that exploits hierarchical dependencies between labels and better handles uncertain samples through label smoothing techniques. In addition, they evaluated several CNN architectures and found that it is difficult for a single model to obtain high and consistent AUC scores for all disease labels simultaneously. To improve prediction accuracy, they applied the Stacking classification technique. The idea of Stacking is to integrate multiple models into a single prediction system to improve prediction accuracy. Specifically, the Stacking model votes on the outputs of all the training networks to obtain the final prediction. In contrast, Devasia et al. [45] developed a multi-label classification method for detecting lung regions on CXR images by using the EfficientNetB4 architecture and a transfer learning approach. Unlike the approach of stacking multiple models to improve prediction performance, transfer learning relies on pre-trained models to adapt to new tasks, enabling efficient feature learning even with limited data. As mentioned above, these

methods rely on encoder-decoders and attention mechanisms to model local image information, ignoring the global co-occurrence relationship between labels.

2.2. Graph convolutional network

In the multi-label classification task, there are much more correlations among labels that we can get by using GCN. Graphs can be constructed based on conditional probability relationships for modeling label dependencies. Specifically, these relationships can be calculated using the conditional probability formula to determine the correlation among labels, which are then represented in a graph format. GCN-based methods [31,32] utilize the feature descriptions and correlation matrix of graph convolutional neural networks to train classifiers. Chen et al. [30] proposed CheXGCN, a novel labeled coevolutionary learning framework based on GCN. The model directly stacks multiple GCN layers and then embeds features from images extracted by the convolutional network to generate a global classifier that explicitly explores multiple correlations among labeled pathologies for the CXR image classification task. Wang et al. [33] proposed a Bi-modal Bridged Graph Convolutional Network (BB-GCN) model for solving problems in multi-label CXR recognition. The model consists of a backbone module, a pathology label co-occurrence relation embedding (LCE) module, and a transformer bridged graph (TBG) module, which can effectively improve the recognition performance and achieve better results. The knowledge enhancement framework proposed by Wu et al. [37] includes a visual encoder and a knowledge encoder, which are used to learn the neural representation of the input image and the medical knowledge graph, respectively. While training the visual encoder, they keep the parameters of the knowledge encoder frozen and learn a set of prompt vectors for efficient adaptation. These studies have proved that for multi-label classification tasks, the frequency of co-occurrence between labels or the conditional probability of occurrence of a category can be used as a priori knowledge to help classification. GCN can model the topological structures of training data by learning the structural similarities among the labels, which is helpful for multi-label classification.

2.3. Attention mechanism

Attention mechanisms are widely used due to their ability to focus on regions of interest and suppress irrelevant background regions by automatically adjusting weights. These mechanisms allow the network to autonomously focus on the parts of interest in the input image, which improves the feature extraction and representation performance. Huang et al. [26] proposed an attention-based framework for learning global and local representations by contrasting image sub-regions and words in the paired report. Kamal et al. [27] proposed Anatomy-XNet, which integrates anatomical knowledge into a deep learning model for attention guidance, thereby improving classification accuracy.

For visual attention mechanisms, the most important thing is how to generate attentional maps that express the priority attention levels for different regions in a single image. So far, there are two techniques to establish feature correlations among different regions. The first one is a self-attention mechanism that captures contextual features of long-range scales in images. The second one is to use large convolutional kernels to establish correlations and generate attention maps. However, large convolutional kernels would significantly increase computational consumption and model parameters. To address this problem, Guo et al. [46] proposed a method to capture long-range relations by decomposing large convolutional kernel operations, which combines the advantages of self-attention and large convolutional kernels. Since some pathologies in chest X-ray images are highly variable in shape and size, and the LKA mechanism can capture larger-scale image contextual feature associations, it is well suited for multi-label disease classification tasks.

3. Method

In this section, we present the proposed multi-label classification model with large-scale image context information and label correlation, as shown in Fig. 1. We first provide an overview of the proposed network structure and introduce each module of the network in detail in the following subsections.

3.1. Overview

The proposed method has three main components: Backbone, Seg-Fusion Block, and GCN Block. Firstly, we pre-trained UNet from a small dataset with organ-level annotations for localizing regions of interest in CXR images. Secondly, we constructed a Backbone network with a four-stage LKA mechanism and superimposed the previous anatomical segmentation results on the feature maps obtained at each stage to obtain a fused feature map. In addition, we use the graph convolutional network as a separate branch to obtain the co-occurrence matrix in all labels from the training dataset. The specific workflow is as follows, the training images are input to the backbone network for multi-scale feature extraction, the feature mask is performed by the pre-trained UNet to obtain the features in the region of interest, and then GAP pooling and aggregation are performed on the features in different stages. For the labeling of the training data, the GCN network is initialized by calculating the co-occurrence matrix and word vectorization for different disease categories, and then the graph relations are converted into feature vectors by a classifier. Finally, the prediction of disease diagnosis is implemented by fusing image semantic features and text correlation features.

3.2. Large kernel attention Block

We build a Backbone network with a four-stage LKA Block, as shown in Fig. 2. The LKA Block can be written as:

$$AttentionMap = Conv_{1\times1}(DWDConv(DWConv(F))) \tag{1}$$

$$FeatureMap = F + AttentionMap \otimes F \tag{2}$$

Here, $F \in R^{C\times H\times W}$ is the input feature matrix. *DWConv* is the depth-wise convolution. *DWDConv* is the depth-wise dilation convolution. $Conv_{1\times1}$ is the 1×1 channel convolution. $AttentionMap \in R^{C\times H\times W}$ denotes the attention map. $FeatureMap \in R^{C\times H\times W}$ denotes the feature map. The value in the attention map indicates the importance of each feature. $\otimes$ means element-wise product.

The LKA mechanism relies on the decomposition operation of a large convolutional kernel. We decompose a large convolution kernel into a depth-wise convolution kernel, a depth-wise dilation convolution kernel, and a 1×1 channel convolution kernel. Specifically, we decompose a convolution kernel of size $K \times K$ into a $(2d\text{-}1) \times (2d\text{-}1)$ depth-wise convolution kernel, a $K/d \times K/d$ depth-wise dilation convolution kernel, and a 1×1 channel convolution kernel, where the size of dilation is d.

After the decomposition operation of the large convolution kernel, the original convolution operation can be represented as three layers of different types of convolution operations. Since this operation decomposes a large convolution kernel into two convolution kernels of different scales, it achieves multi-scale feature extraction which corresponds to diseases in smaller and larger regions of the image representation, respectively. Compared to single-scale convolution kernels, this operation can both obtain more feature information and also capture feature relationships over longer distances. Similar to depth-separable convolution, the number of parameters of the model is greatly reduced after the decomposition operation, and the purpose of capturing long-distance feature relationships in image context information is achieved with less computational cost.

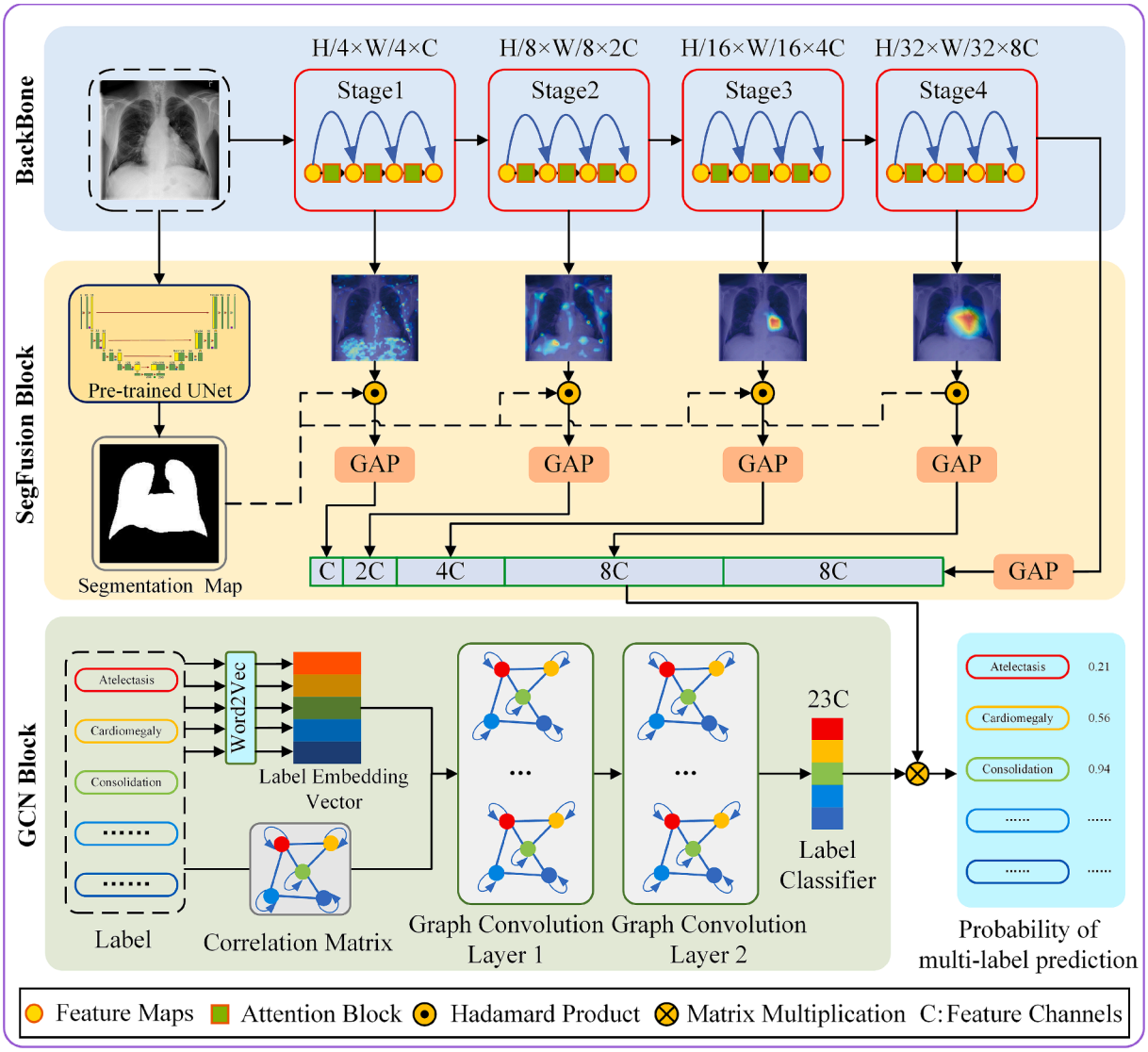


Fig. 1. The overall architecture of our method. The Backbone network consists of a four-stage Large Kernel Attention Block that extracts a larger range of image contextual features. The SegFusion Block mainly contains a pre-trained UNet for focusing on the lung region in the input image. The GCN Block works as a separate branch that obtains the co-occurrence relation matrix of all the training labels.

3.3. SegFusion Block

To enhance the ability to focus on the lung and heart regions in the feature map, we consider overlaying the anatomical segmentation map with the feature map to obtain the fused feature map. By this operation, the pixel values in the feature map corresponding to the regions of interest in the anatomical segmentation map will be enhanced, while the pixel values in other regions will be suppressed. This module is shown in Fig. 3.

To obtain the anatomical segmentation map, we train the UNet model on a subset of the JSRT dataset and segment both the lung region and the heart region by merging the labeled data. The resulting pre-trained model is used to obtain the image segmentation maps for feature filtering and fusion. Specifically, we superimpose the anatomical segmentation maps onto the feature maps obtained by convolution at each stage of the backbone network to obtain the segmentation fusion feature maps. The operation can be expressed as follows:

$$SFM_i = Resize(SegmentationMap) \odot Resize(FeatureMap_i) \tag{3}$$

$$Output_i = GAP(SFM_i) \tag{4}$$

$$Output = Concat(Output_i, GAP(FeatureMap_4)) \tag{5}$$

Since the feature maps of each stage have different dimensions, and they are not the same size as the segmentation maps predicted by the UNet model. We consider scaling both the feature map and the segmentation map to the same size, and then perform dot product on them to obtain the segmentation fusion feature map. The same operation is performed on each of the four stages of the feature maps to obtain the segmentation fusion feature maps at four different scales. To fuse the feature maps at different scales, we use the GAP operation, which compresses the shape size to a single pixel and keeps the number of channels constant, to obtain the GAP feature map. Also, to avoid the limitation of over-focusing on the lung and heart regions that may be brought by the fusion segmentation map operation, we obtain the global GAP feature maps by directly performing GAP operations on the final output feature maps of the backbone network. Finally, we concatenate these GAP feature maps for the combination with the classifier output from the GCN network later.

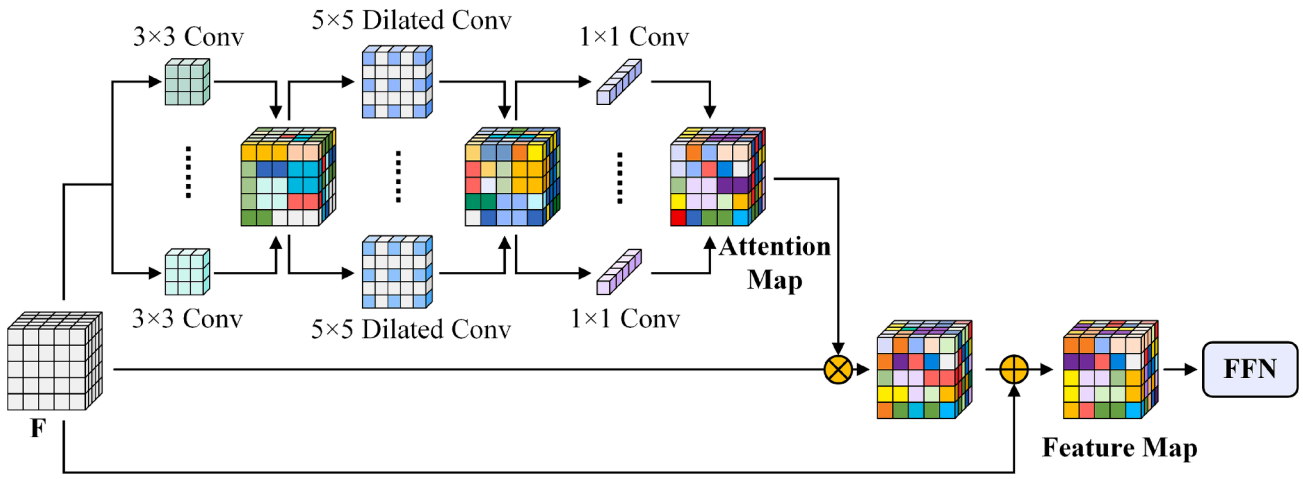


Fig. 2. Large Kernel Attention (LKA) Block. The LKA mechanism relies on the decomposition operation of a large convolution kernel, which decomposes a large convolution kernel into a depth convolution kernel, a depth dilation convolution kernel, and a 1 × 1 channel convolution kernel. The input features are used to generate an attention map by LKA, which is then superimposed on the original features by tensor multiplication. To preserve the initial feature information and relieve the gradient vanishing problem in the deep network, the input feature maps are finally fused with the feature maps superimposed on the attention maps through skip connections.

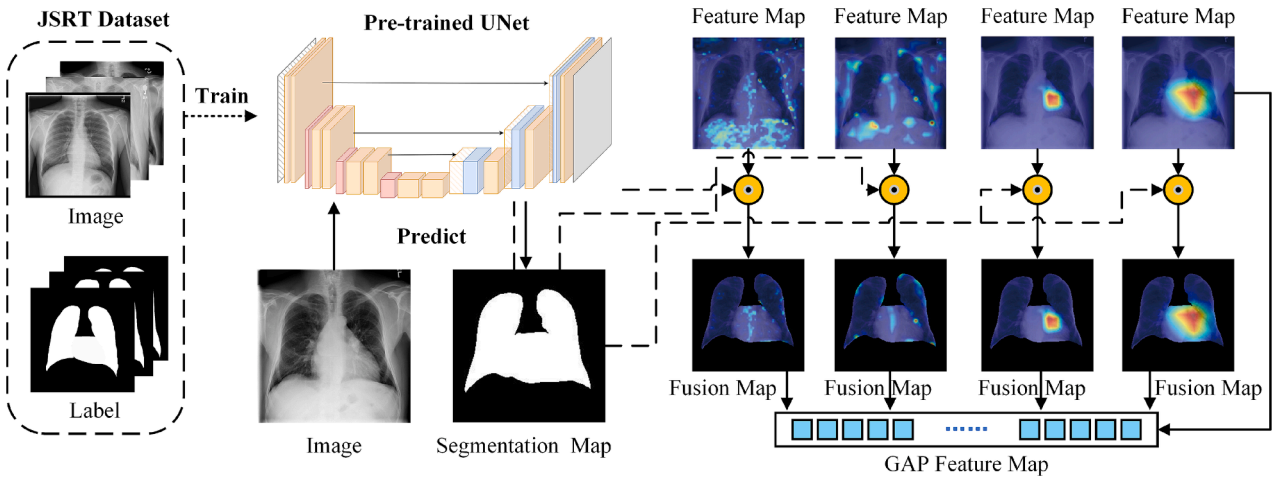


Fig. 3. SegFusion Block. The lung and heart regions of the input image are segmented by pre-training the UNet model on the JSRT dataset. Then, the feature maps of each stage of the backbone network are filtered and fused based on these segmented regions. To avoid losing information outside the segmented regions, we retain both the original backbone network output features in the fused GAP feature map.

3.4. GCN Block

The basic idea of the graphical convolutional neural network is to update the node representation by propagating between nodes to realize the label relevance training process. We use $H^l \in \mathbb{R}^{n\times d}$ to denote the node features of a layer and use $A \in \mathbb{R}^{n\times n}$ to denote the correlation matrix, then the node features of the next layer are $H^{l+1} \in \mathbb{R}^{n\times d'}$, where n denotes the number of nodes or categories, d denotes the node dimension of the layer, and d' denotes the node dimension of the next layer. The main purpose of the GCN is to learn the transformation function $f(\cdot)$ between each layer, i.e.:

$$H^{l+1} = f(H^l, A) \tag{6}$$

After the convolution of this layer, the transform function can be expressed as:

$$H^{l+1} = h(\widehat{A}H^lG^l) \tag{7}$$

where, G^l is the transformation matrix obtained by training with dimension $d \times d'$, $\widehat{A}$ is the normalized matrix of the correlation matrix A, and $h(\cdot)$ is the nonlinear activation function.

In GCN, the correlation matrix A is usually predefined, however, when the dataset contains amounts of different types of nodes by using the matrix would not capture the similarity among nodes effectively. To address this problem, the conditional probability matrix is used as the correlation matrix in this paper. Converting the category labels to be classified into the word embedding vectors is a common approach that can make it easier for the model to learn the correlation between the labels. Here, we use the Word2Vec model from the field of natural language processing to obtain a word embedding vector representation for each category.

We input the encoded label word vectors and correlation matrix into the constructed GCN network to learn the co-occurrence relationship of labels. Finally, we multiply the label classifier of the GCN model with the GAP feature map of the SegFusion Block, and then obtain the classification prediction probability of each label.

3.5. Network implementation details

Our backbone network structure consists of four stages with

decreasing resolutions of the output spaces, namely H/4 × W/4, H/8 × W/8, H/16 × W/16, and H/32 × W/32. H and W denote the height and width of the input image. We use a common processing method to increase the number of output channels while reducing the resolution. For each stage, we first pool the input, then all other layers throughout the stage maintain the same spatial resolutions and the number of channels. The backbone network parameter settings are shown in Table 1.

For the large kernel separable convolution for obtaining the attention maps, we decompose the large kernel of size K = 21 into 5 × 5 depth-wise convolution, 7 × 7 depth-wise dilation convolution, and 1 × 1 depth-wise convolution. For depth-wise dilation convolution select dilation d = 3. We use two Graph Convolution Layers to build the GCN for modeling data correlation. Because we tested that even adding more Graph Convolution Layers on the dataset we used has no improvement for data correlation learning. We use Focal Loss as the loss function of our network and train it by an Adam optimization function with standard parameters, and the learning rate is fixed at 0.0001.

4. Experiment

In this section, we conduct comparative experiments with existing classical models and other state-of-the-art methods on three CXR datasets to demonstrate the effectiveness and efficiency of our method.

4.1. Datasets

CheXpert: CheXpert (Chest eXpert) consists of 224,316 CXR from 65,240 patients, and each data set in the dataset contains at least one frontal view image labeled as CXR observations for the presence of 14 common diseases. The validation and test sets consisted of 234 CXR from 200 patients and 668 CXR from 500 patients, respectively. We divide the data set according to the official one for experimentation and fair comparison with previous work.

NIH: The NIH CXR dataset consists of 112,120 X-rays from 30,805 unique patients across 14 different diseases. We followed the official NIH divisions of 70 % for training, 10 % for validation, and 20 % for testing to conduct experiments and make a fair comparison with previous work.

MIMIC-CXR-JPG: The MIMIC-CXR-JPG dataset contains 377,110 images in JPG format and labels corresponding to 14 categories, derived from 227,827 free-text radiology reports associated with these images. We divide the data set according to the official one for experimentation and fair comparison with previous work.

JSRT: We use the JSRT as annotated dataset to train the UNet model. The segmentation annotations for JSRT, including heart and lung, are obtained from.

The percentage of label distribution for the 14 diseases on the CheXpert dataset, NIH dataset, and MIMIC-CXR-JPG dataset are shown in Fig. 4.

As shown in Fig. 5, we obtained many disease label pairs with co-occurrence relationships by counting the number of occurrences of disease labels for all classification targets in the three datasets respectively. For example, pulmonary disease is often associated with pleural effusion, and atelectasis is often co-occurring with infiltration. This phenomenon provides preliminary evidence for the feasibility of using the GCN module to learn correlations between labels.

Table 1
The Backbone Network Parameter Settings.

Stage	Block Number	Out Size	Channel Number	FFN Expansion Ratio
1	3	H/4 × W/4	64	8
2	3	H/8 × W/8	128	8
3	5	H/16 × W/16	256	4
4	2	H/32 × W/32	512	4

*H and W denote the height and width of the input image.

4.2. Data preprocessing

We train classification models in the frontal views of images in each dataset referred. We evaluated the CheXpert dataset for 5 categories of diseases: Atelectasis, Cardiomegaly, Consolidation, Edema, and Pleural Effusion. For uncertain labels included in the CheXpert dataset (value of −1), we consider them as the existing disease (value of 1). We evaluated the NIH dataset for 14 categories of disease except for No Finding labels. We evaluated the MIMIC-CXR-JPG dataset for all 13 disease categories as well as the No Finding labels. For uncertain labels in the MIMIC-CXR-JPG dataset (value of −1), we consider them did not exist (value of 0).

We resized the CXR images to 224 × 224 pixels as input and enhanced the data using basic random rotation, translation, and scaling operations to improve the generalization capability of the model.

4.3. Evaluation metric

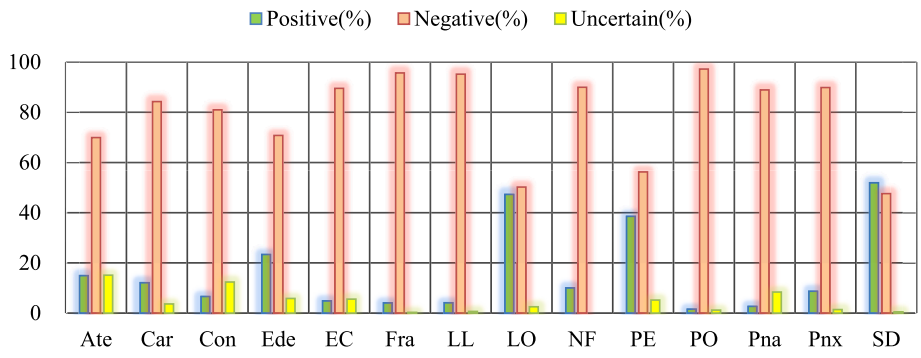
To comprehensively evaluate the classification performance of the model, the area under the ROC curve (AUC) was used to evaluate the model. The horizontal coordinate of the ROC curve is the False Positive Rate (FPR), defined as $FPR = FP/(FP + TN)$, which represents the proportion of all negative samples that were incorrectly considered as positive; the vertical coordinate is the Ture Positive Rate (TPR), defined as $TPR = TP/(TP + FN)$, which represents the proportion of all positive samples that were correctly considered as positive. The final quantitative evaluation of the model was performed by using the average AUC of all labels as the evaluation index, and the AUC was closer to 1, the better the classification performance of the model.

4.4. Ablation studies

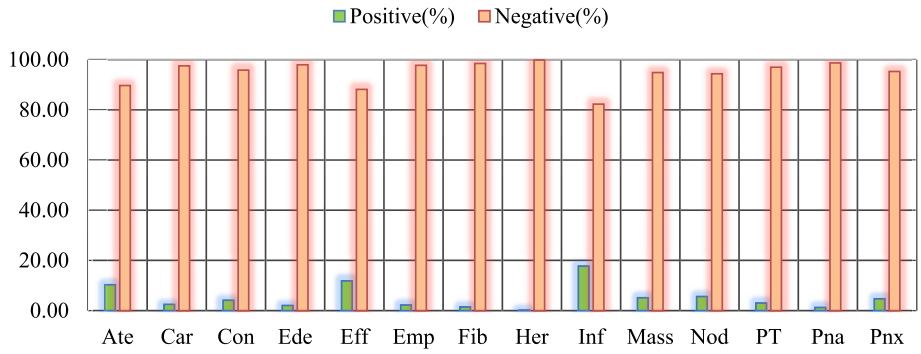
We performed five groups of ablation studies to evaluate the specific effects of each part of the network. Experiment 1 used DenseNet121, which is popular among recent studies, to achieve the goal of multi-label classification after feature extraction. Experiment 2 uses a four-stage network constructed by LKA Block as the backbone network for feature extraction. The first two sets of experiments are mainly used to verify the enhancement of the LKA mechanism on multi-label classification results. Experiment 3 is based on Experiment 2 and adds a model with an anatomical segmentation fusion mechanism (Backbone + SegFusion). Experiment 4 is based on Experiment 2 and adds a model with a label relationship learning module (Backbone + GCN). Experiment 5 is based on Experiment 2 and adds a model with both modules (Backbone + SegFusion + GCN).

The results in Table 2, Table 3, and Table 4 show that the classification effect of Experiment 2 is significantly better than that of Experiment 1, which proves that the feature extraction block by adding the attention mechanism is more accurate than the traditional convolutional approach. Experiment 3 (Backbone + SegFusion) and Experiment 4 (Backbone + GCN) show more advantages than Experiment 2 (Backone), among which Experiment 5 (Backbone + SegFusion + GCN) achieves the best results, and it proved that the attention mechanism based on fused anatomical segmentation and label co-occurrence relations in combination significantly improve the performance of disease classification.

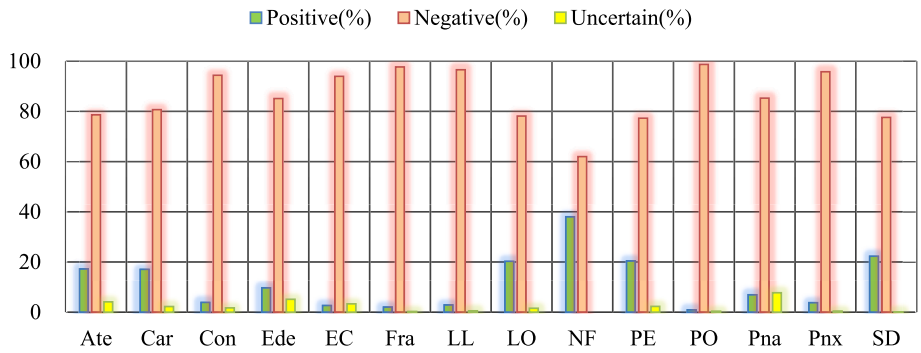
The results of the ablation studies demonstrate that three operations can improve prediction results. The first is the introduction of the attention mechanism, the second is to guide the attention map by fusing anatomical segmentation, and the third is to introduce the correlation of labels. Since LKA Block decomposes a large convolution kernel into two scales, it achieves the goal of multi-scale feature extraction, targeting diseases in smaller and larger regions of image representation,



(a) Percentage of label distribution for the 14 diseases on the CheXpert dataset.



(b) Percentage of label distribution for the 14 diseases on the NIH dataset.

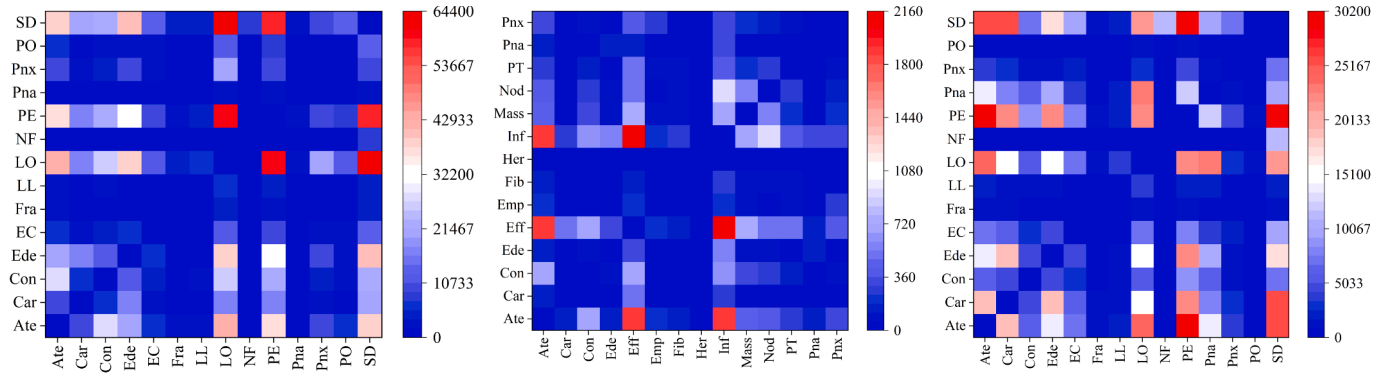


(c) Percentage of label distribution for the 14 diseases on the MIMIC-CXR-JPG dataset.

Fig. 4. Percentage of label distribution for the 14 diseases on the three datasets. The abbreviations for the diseases in the figure are as follows: Atelectasis(Ate), Cardiomegaly(Car), Consolidation(Con), Edema(Ede), Effusion (Eff), Pleural Effusion(PE), Emphysema (Emp), Fibrosis (Fib), Hernia (Her), Infiltration (Inf), Mass, Nodule (Nod), Pleural_Thickening (PT), Pneumonia (Pna), and Pneumothorax (Pnx), Enlarged Cardiomediastinum(EC), Fracture (Fra), Lung Lesion (LL), Lung Opacity (LO), Pleural Other (PO), Support Devices (SD), No Finding (NF).

respectively. Compared with the single-scale convolution kernel, much more featured information could be obtained, and the overall classification effect is significantly improved. By incorporating anatomical information into the attention map, the model can be guided to focus on learning key regions relevant to the classification task. This enhanced attention mechanism strengthens the ability of the model's perception and application. Considering label relevance can provide more rich information and context, and it will help reduce the error rate of model classification and imbalance the differences among labels.

We also performed ablation experiments for convolutional kernel size in LKA. From the results of the ablation experiments in Table 5, it can be seen that the effect of different convolution kernel sizes in LKA on the model performance is very significant. In the table, DW-Conv denotes the deep convolution which can utilize the local context information of the image. DW-D-Conv denotes the deep dilation convolution which plays a role in capturing the long-range dependency of the image. Dilation denotes the dilation rate of DW-D-Conv. As can be seen from the table, the average AUC values of the model on the three datasets gradually increase as the convolution kernel size and dilation rate gradually increase. Especially on the CheXpert and MIMIC-CXR-JPG datasets, the AUC shows a large increase, which is even close to 1 % initially. This is due to the larger convolutional kernel being able to capture a wider range of contextual information, making the model more suitable for handling the complex patterns of CXR images. Notably, while larger convolutional kernel sizes result in better performance, the improvement from 21 $\times$ 21 to 28 $\times$ 28 becomes very small. The AUC on the CheXpert and MIMIC-CXR-JPG datasets is only improved by 0.1 %, and even not on the NIH dataset. This shows that the benefits are decreasing as the size of the convolution kernel exceeds a certain threshold. The reason for this is that feature extraction is already sufficient, and at this time, if we choose a too-large convolutional kernel instead, it will increase the computational complexity and the difficulty of model



(a) CheXpert (b) NIH (c) MIMIC-CXR-JPG

Fig. 5. Illustration of label co-occurrence matrixes extracted from the CheXpert (a), NIH (b), and MIMIC-CXR (c) datasets, respectively. The abbreviations for the diseases in the figure are as follows: Atelectasis(Ate), Cardiomegaly(Car), Consolidation(Con), Edema(Ede), Effusion (Eff), Pleural Effusion(PE), Emphysema (Emp), Fibrosis (Fib), Hernia (Her), Infiltration (Inf), Mass, Nodule (Nod), Pleural_Thickening (PT), Pneumonia (Pna), and Pneumothorax (Pnx), Enlarged Cardiomediastinum(EC), Fracture (Fra), Lung Lesion (LL), Lung Opacity (LO), Pleural Other (PO), Support Devices (SD), No Finding (NF).

Table 2
Comparison Of Classification AUC For Ablation Studies On The CheXpert Dataset.

Method	Atelectasis	Cardiomegaly	Consolidation	Edema	Pleural Effusion	Average AUC
DenseNet121	0.830	0.887	0.885	0.892	0.911	0.881
Backbone	0.852	0.901	0.890	0.912	0.920	0.895
Backbone + SegFusion	0.835	0.900	0.894	0.921	0.957	0.901
Backbone + GCN	0.841	0.905	0.892	0.919	0.960	0.903
Backbone + SegFusion + GCN	0.866	0.907	0.920	0.920	0.962	0.915

training.

In the work of Chen et al. [47] it is proposed to binarize the label correlation to avoid noise and overfitting problems, and filter the noise using a threshold τ. At the same time, they adjusted the feature weights of the node itself and neighboring nodes by the parameter p. It prevents over-smoothing of features and thus balances the effect of node information on feature updates. Our GCN network adopts the same strategy, and we analyze the effects of these parameters through ablation experiments.

As can be seen from Table 6, the average AUC value of the model on all datasets shows a decreasing trend as the number of GCN layers increases. The shallower GCN network (2 layers) achieves the best performance, and as the number of layers increases, the AUC of the model gradually decreases. A known problem with GCN networks is that as the number of layers increases, the features of the nodes may gradually become too similar, which is known as the phenomenon of over-smoothing. This can make it difficult for the model to distinguish the features of different nodes, which as a result reduces the model's performance. The results of the ablation experiments show that a shallower GCN is sufficient to extract the graph structure information in this task and that increasing the number of layers does not improve its performance.

As shown in Fig. 6a, when small probabilistic noise edges are gradually filtered out as the value of τ becomes larger, the multi-label recognition accuracy is improved. However, when too many edges are filtered out, the accuracy decreases because the relevant neighbors are also ignored. If no edges are filtered, the model will not converge. Therefore, there is no result for $\tau = 0$ in this figure. From the figure, we know that the optimal value of τ is 0.3. As shown in Fig. 6b, this figure shows the importance of balancing the weights between the node itself and its neighbors when updating the node features in the GCN. We can see that it achieves the best performance on all three datasets when p is around 0.2 to 0.3. If p is too small, the nodes of the graph cannot get enough relevant information from neighboring nodes. If p is too large, it will lead to excessive smoothing that cannot converge. In this paper, the optimal value of p is chosen to be 0.2.

4.5. Results and analysis

To demonstrate the effectiveness of the proposed method, we conduct existing experiments on our method compared with other state-of-the-art methods. Table 7, Table 8, and Table 9 compared our proposed model with other methods referred to.

4.5.1 Performance on the CheXpert dataset

Table 7 shows the classification results of eight different methods trained on the CheXpert dataset. The results in Table 7 show that our model achieves an average AUC of 0.915 on the classification task for five diseases, ahead of most current methods. Among them, the classification AUC exceeded 0.9 for all four diseases, Cardiomegaly, Consolidation, Edema, and Pleural Effusion. The method in this paper is slightly lower than Pham's method (0.940) and the same as Kamal's (0.915) in average AUC (0.915), but better than the majority of other methods. The Pleural Effusion class has an AUC of 0.962, which is about 3 % higher than the other methods in the table. Fig. 8(a) illustrates the ROC curves of our method over the 5 pathologies on CheXpert. The ROC curves for different pathologies show that certain pathologies are better categorized than others. For example, the curve for Pleural Effusion is almost near the upper left corner, indicating a high rate of true positives and a low rate of false positives. The curve for Atelectasis, on the other hand, is relatively low, indicating a relatively weak classification performance for this condition. However, its AUC also reached 0.866, which is higher than the average of the other methods.

4.5.2 Performance on the NIH dataset

Table 8 compared the results of our proposed method with eleven different previously published methods on the NIH dataset. The results in Table 8 show that our model achieves an average AUC of 0.842 on the

Table 3
Comparison Of Classification AUC For Ablation Studies On The NIH Dataset.

Method	Ate	Car	Con	Ede	Eff	Emp	Fib	Her	Inf	Mass	Nod	PT	Pna	Pnx	Average AUC
DenseNet121	0.777	0.883	0.748	0.862	0.829	0.909	0.847	0.868	0.710	0.826	0.755	0.769	0.732	0.861	0.813
Backbone	0.786	0.901	0.760	0.871	0.838	0.923	0.855	0.888	0.709	0.845	0.788	0.808	0.744	0.889	0.829
Backbone + SegFusion	0.790	0.906	0.771	0.884	0.845	0.935	0.866	0.895	0.711	0.850	0.793	0.815	0.749	0.890	0.836
Backbone + GCN	0.789	0.911	0.779	0.879	0.851	0.936	0.861	0.903	0.718	0.853	0.791	0.798	0.753	0.902	0.837
Backbone + SegFusion + GCN	0.800	0.913	0.778	0.880	0.848	0.936	0.869	0.910	0.719	0.860	0.795	0.814	0.755	0.903	0.842

The 14 findings for NIH datasets are Atelectasis(Ate), Cardiomegaly (Car), Consolidation (Con), Edema (Ede), Effusion (Eff), Emphysema (Emp), Fibrosis (Fib), Hernia (Her), Infiltration (Inf), Mass, Nodule (Nod), Pleural_Thickening (PT), Pneumonia (Pna), and Pneumothorax (Pnx).

Table 4
Comparison Of Classification AUC For Ablation Studies On The MIMIC-CXR-JPG Dataset.

Method	Ate	Car	Con	Ede	EC	Fra	LL	LO	PE	Pna	Pnx	PO	SD	NF	Average AUC
DenseNet121	0.758	0.777	0.758	0.822	0.779	0.755	0.741	0.701	0.878	0.752	0.884	0.888	0.891	0.801	0.799
Backbone	0.771	0.790	0.782	0.830	0.783	0.772	0.756	0.716	0.887	0.760	0.885	0.887	0.900	0.812	0.809
Backbone + SegFusion	0.770	0.790	0.786	0.835	0.789	0.784	0.755	0.730	0.890	0.763	0.900	0.906	0.905	0.819	0.816
Backbone + GCN	0.774	0.792	0.784	0.833	0.791	0.800	0.760	0.723	0.878	0.767	0.897	0.910	0.910	0.820	0.817
Backbone + SegFusion + GCN	0.779	0.804	0.791	0.845	0.801	0.803	0.766	0.732	0.894	0.770	0.901	0.915	0.917	0.830	0.825

The 14 pathologies for the MIMIC-CXR-JPG datasets are Atelectasis (Ate), Cardiomegaly (Car), Consolidation (Con), Edema (Ede), Enlarged Cardiomediastinum(EC), Fracture (Fra), Lung Lesion (LL), Lung Opacity (LO), Pleural Effusion (PE), Pneumonia (Pna), Pneumothorax(Pnx), Pleural Other (P.O.), Support Devices (S.D.), No Finding (NF).

Table 5
Ablation study of different kernel sizes in LKA.

Kernel Size	DW-Conv	DW-D-Conv	Dilation	Average AUC CheXpert	NIH	MIMIC-CXR-JPG
7 × 7	3 × 3	3 × 3	2	0.898	0.831	0.812
14 × 14	4 × 4	5 × 5	3	0.909	0.838	0.820
21 × 21	5 × 5	7 × 7	3	0.915	0.842	0.825
28 × 28	7 × 7	7 × 7	4	0.916	0.842	0.826

Table 6
Ablation study of different layers of GCN.

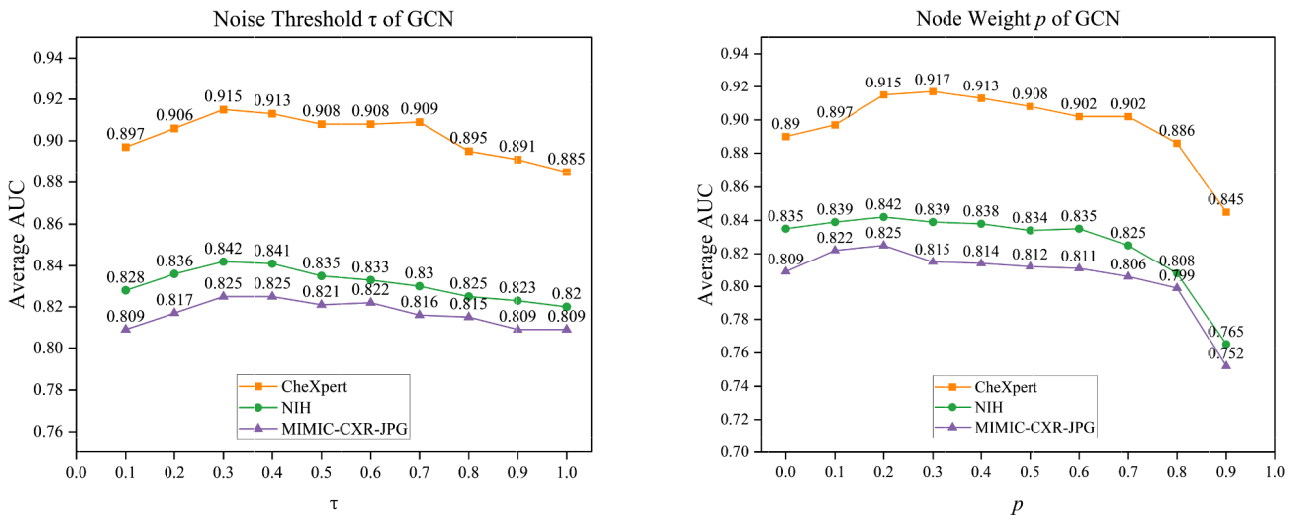
GCN Layers	Average AUC		
	CheXpert	NIH	MIMIC-CXR-JPG
2	0.915	0.842	0.825
3	0.910	0.838	0.820
4	0.901	0.830	0.816

classification task of fourteen diseases, which exceeds most current methods. Although there is a slight gap with the state-of-the-art methods (i.e. Kamal's 0.851, CvTGNet's 0.847), it has been significantly improved compared to the earlier methods. Among them, the classification AUC for three diseases, Fibrosis, Mass, and Pleural_Thickening, exceeded existing methods, and the classification AUC for four diseases, Cardiomegaly, Emphysema, Hernia, and Pneumothorax, all exceeded 0.9. Fig. 8 (b) illustrates the ROC curves of our method over the 14 pathologies on NIH. The ROC curves for each pathology also corroborate this result more clearly, with the ROC curves for Emphysema, Cardiomegaly, and Hernia being closer to the upper left corner, suggesting that these pathologies are better categorized. In contrast, the curves for Infiltration, Fibrosis, and Pneumonia are weaker, indicating that the model is less able to differentiate on these conditions.

4.5.3 Performance on the MIMIC-CXR-JPG dataset

Table 9 compared the results of our proposed method with the six previously published methods on the MIMIC-CXR-JPG dataset. The results in Table 9 show that our model achieves an average AUC of 0.825 on thirteen diseases as well as on the classification task considering uncertainty, exceeding most current methods. Among them, the classification AUC for four diseases, Enlarged Cardiomediastinum, Fracture, Pneumonia, and Pleural Other, exceeded the existing methods, and for three diseases, Pneumothorax, Pleural Other, and Support Devices classification AUCs exceeded 0.9 for all three diseases. The method in this paper performed particularly well on pneumonia (PE) and pleural effusion (PO), with AUCs of 0.894 for PE and 0.915 for PO, both higher than most other methods. Fig. 8(c) illustrates the ROC curves of our method over the 14 pathologies on MIMIC-CXR-JPG.

In Fig. 7, we compare the different performance metrics of several state-of-the-art methods. From the figure, we can find that the parameter numbers of this paper's method are 14.5 M, which is significantly lower than the other compared models, indicating a more lightweight design. Despite its FLOPs being 18.4G, which is slightly higher than that of



(a) Noise Threshold τ of GCN (b) Node Weight p of GCN.

Fig. 6. Average AUC performance on the three datasets for different GCN parameter values.

Table 7
Pathology-Wise Performance Comparison Of The Proposed Method With State-Of-The-Art Systems On The CheXpert Dataset.

Method	Atelectasis	Cardiomegaly	Consolidation	Edema	Pleural Effusion	Average AUC
Allaouzi [48]	0.720	0.880	0.870	0.770	0.900	0.828
Irvin [23]	0.858	0.832	0.941	0.899	0.934	0.893
Bajwa [49]	0.833	0.815	0.853	0.914	0.908	0.865
Chen B [30]	0.736	0.877	0.784	0.886	0.907	0.838
Anatomy-XNet [27]	0.866	0.879	0.953	0.931	0.947	0.915
Pham [44]	0.909	0.910	0.957	0.958	0.964	0.940
Suxia Xing [50]	0.705	0.856	0.751	0.907	0.897	0.823
MANet [51]	0.814	0.866	0.922	0.916	0.899	0.883
CvTGNet [52]	0.811	0.922	0.844	0.895	0.925	0.879
ADNet [53]	0.867	0.869	0.920	0.941	0.934	0.906
Ours	0.866	0.907	0.920	0.920	**0.962**	0.915

Table 8
Pathology-Wise Performance Comparison Of The Proposed Method With State-Of-The-Art Systems On The NIH Dataset.

Method	Ate	Car	Con	Ede	Eff	Emp	Fib	Her	Inf	Mass	Nod	PT	Pna	Pnx	Average AUC
Ho [54]	0.795	0.887	0.786	0.892	0.875	0.875	0.756	0.836	0.703	0.835	0.716	0.774	0.742	0.863	0.810
CRAL [55]	0.781	0.880	0.754	0.850	0.829	0.908	0.830	0.917	0.702	0.834	0.773	0.778	0.729	0.857	0.816
Rajpurkar [22]	0.780	0.882	0.754	0.850	0.827	0.925	0.822	0.932	0.689	0.831	0.781	0.793	0.735	0.851	0.818
DualCheXNet [56]	0.784	0.888	0.746	0.852	0.831	0.942	0.837	0.912	0.705	0.838	0.796	0.796	0.727	0.876	0.823
LLAGNet [57]	0.783	0.885	0.754	0.851	0.834	0.939	0.832	0.916	0.703	0.841	0.790	0.798	0.729	0.877	0.824
Wang [33]	0.779	0.895	0.759	0.855	0.836	0.933	0.838	0.938	0.710	0.834	0.777	0.791	0.737	0.878	0.826
Yan [58]	0.792	0.881	0.760	0.847	0.842	0.942	0.833	0.934	0.710	0.847	0.811	0.808	0.740	0.876	0.830
Luo [59]	0.789	0.907	0.768	0.861	0.842	0.940	0.838	0.937	0.718	0.838	0.799	0.804	0.742	0.906	0.835
MANet [51]	0.814	0.894	0.802	0.886	0.863	0.852	0.828	0.921	0.700	0.834	0.778	0.768	0.753	0.855	0.825
Keidar [60]	0.806	0.909	0.804	0.892	0.869	0.909	0.815	0.918	0.706	0.839	0.771	0.780	0.765	0.855	0.831
Anatomy-XNet [27]	0.831	0.914	0.809	0.899	0.886	0.929	0.844	0.964	0.717	0.860	0.804	0.798	0.771	0.882	0.851
CvTGNet [52]	0.816	0.927	0.818	0.927	0.900	0.879	0.829	0.962	0.706	0.845	0.755	0.809	0.796	0.877	0.847
ADNet [53]	0.801	0.891	0.769	0.865	0.843	0.942	0.853	0.926	0.717	0.849	0.822	0.808	0.750	0.897	0.838
Ours	0.800	0.913	0.778	0.880	0.848	0.936	**0.869**	0.910	0.719	**0.860**	0.795	**0.814**	0.755	0.903	0.842

The 14 findings for NIH datasets are Atelectasis(Ate), Cardiomegaly (Car), Consolidation (Con), Edema (Ede), Effusion (Eff), Emphysema (Emp), Fibrosis (Fib), Hernia (Her), Infiltration (Inf), Mass, Nodule (Nod), Pleural_Thickening (PT), Pneumonia (Pna), and Pneumothorax (Pnx).

Table 9
Pathology-Wise Performance Comparison Of The Proposed Method With State-Of-The-Art Systems On The MIMIC-CXR-JPG Dataset.

Method	Ate	Car	Con	Ede	EC	Fra	LL	LO	PE	Pna	Pnx	PO	SD	NF	Average AUC
DN-KG [61]	0.694	0.746	0.640	0.790	0.651	0.605	0.574	0.609	0.809	0.572	0.689	0.650	0.781	0.778	0.685
VSE-GCN [62]	0.722	0.730	0.728	0.799	0.767	0.560	0.623	0.654	0.863	0.588	0.797	0.653	0.789	0.817	0.721
Chexclusion [63]	0.837	0.828	0.844	0.904	0.757	0.718	0.772	0.782	0.933	0.748	0.903	0.848	0.927	0.868	0.834
MANet [51]	0.828	0.819	0.837	0.900	0.745	0.696	0.754	0.772	0.915	0.730	0.880	0.831	0.902	0.859	0.819
Keidar [60]	0.832	0.826	0.842	0.904	0.747	0.713	0.767	0.777	0.929	0.745	0.897	0.842	0.921	0.864	0.829
Anatomy-XNet [27]	0.838	0.827	0.853	0.908	0.755	0.743	0.771	0.788	0.934	0.760	0.909	0.866	0.928	0.869	0.839
CvTGNet [52]	0.830	0.831	0.853	0.893	0.766	0.753	0.758	0.774	0.904	0.751	0.902	0.867	0.915	0.855	0.832
ADNet [53]	0.840	0.840	0.862	0.910	0.771	0.787	0.807	0.792	0.936	0.771	0.930	0.885	0.936	0.879	0.853
Ours	0.779	0.804	0.791	0.845	**0.801**	**0.803**	0.766	0.732	0.894	0.770	0.901	**0.915**	0.917	0.830	0.825

The 14 pathologies for the MIMIC-CXR-JPG datasets are Atelectasis (Ate), Cardiomegaly (Car), Consolidation (Con), Edema (Ede), Enlarged Cardiomediastinum(EC), Fracture (Fra), Lung Lesion (LL), Lung Opacity (LO), Pleural Effusion (PE), Pneumonia (Pna), Pneumothorax(Pnx), Pleural Other (PO), Support Devices (SD), No Finding (NF).

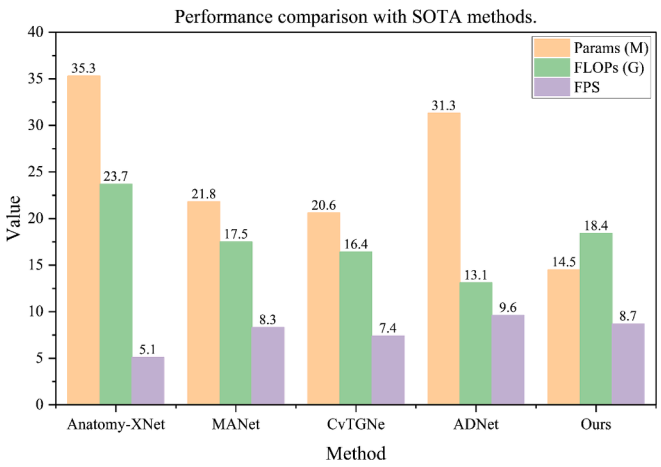


Fig. 7. Performance comparison with state-of-the-art methods.

CvTGNet (16.4G) and close to that of MANet (17.5G), its inference speed still achieves 8.7 FPS, which is the same as that of MANet, and better than that of CvTGNet (7.4 FPS). Compared with ADNet, the parameter volume of this paper's method is reduced by about 53.7 % (14.5 M vs. 31.3 M), significantly reducing the model's complexity and storage requirements while keeping the approximate inference performance. In conclusion, the method in this paper achieves a relatively better balance between the parameter number, computational complexity, and inference speed, considering both performance and resource efficiency.

In our experiments, category-specific heatmaps are generated by using Grad-CAM [36], as shown in Fig. 9. A single image may contain information related to multiple categories, it becomes much more important to obtain category-specific feature maps. Red and yellow areas represent where the model's attention is most focused, and blue and green represent areas of relatively less attention. For example, for Cardiomegaly, the red region is focused on the heart region, indicating that the model correctly identified the location of the pathology. For No Finding, the model generates a wider range of attention maps, possibly because the model has more difficulty recognizing non-pathologic radiographs. With these heat maps, doctors can see exactly what areas the model "focuses" on the most and can explain why the model makes particular predictions. This is particularly important in medical image analysis, which can increase trust in AI models.

4.6. Limitations

The method in this paper, despite achieving excellent results on several publicly available datasets, still has some limitations and shortcomings, mainly focusing on the following aspects:

4.6.1 Long-tailed class distribution problems

There is a serious long-tailed class distribution phenomenon in the NIH and MIMIC-CXR-JPG datasets, which is shown that a few conditions (e.g., common pneumonia, cardiac hypertrophy) have a large proportion of the training data, while some rare conditions have a small sample size. This data imbalance causes the model to be more sensitive to classes with larger sample sizes, and is prone to ignore or misclassify rare conditions with smaller sample sizes. To address this issue, future research could try to introduce data enhancement techniques, resampling strategies, or loss function adjustments (e.g., weighted cross-entropy, loss of focus) to deal with long-tailed distributions. In addition, methods such as generative adversarial networks (GANs) or self-

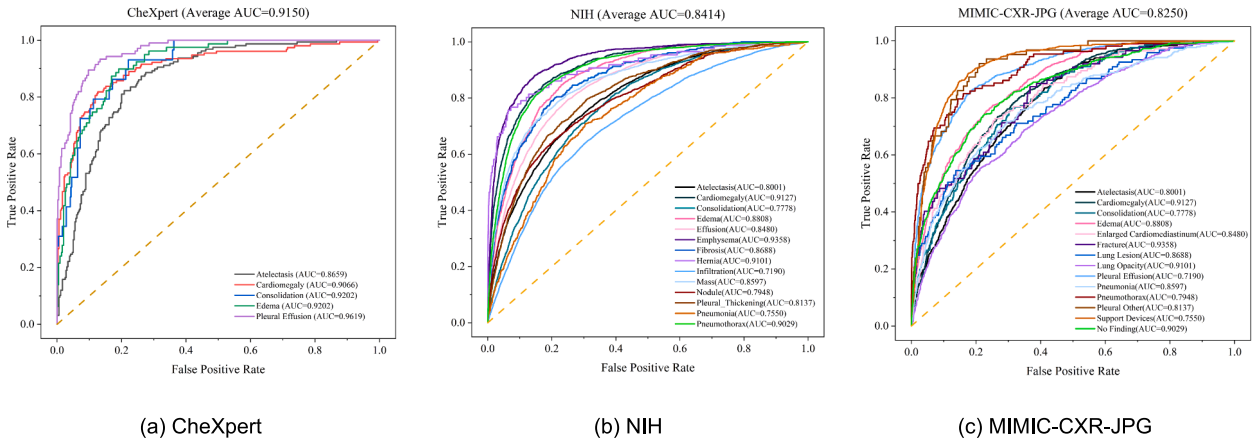


Fig. 8. ROC curve of our method on the CheXpert (a), NIH (b), and MIMIC-CXR-JPG (c) datasets.

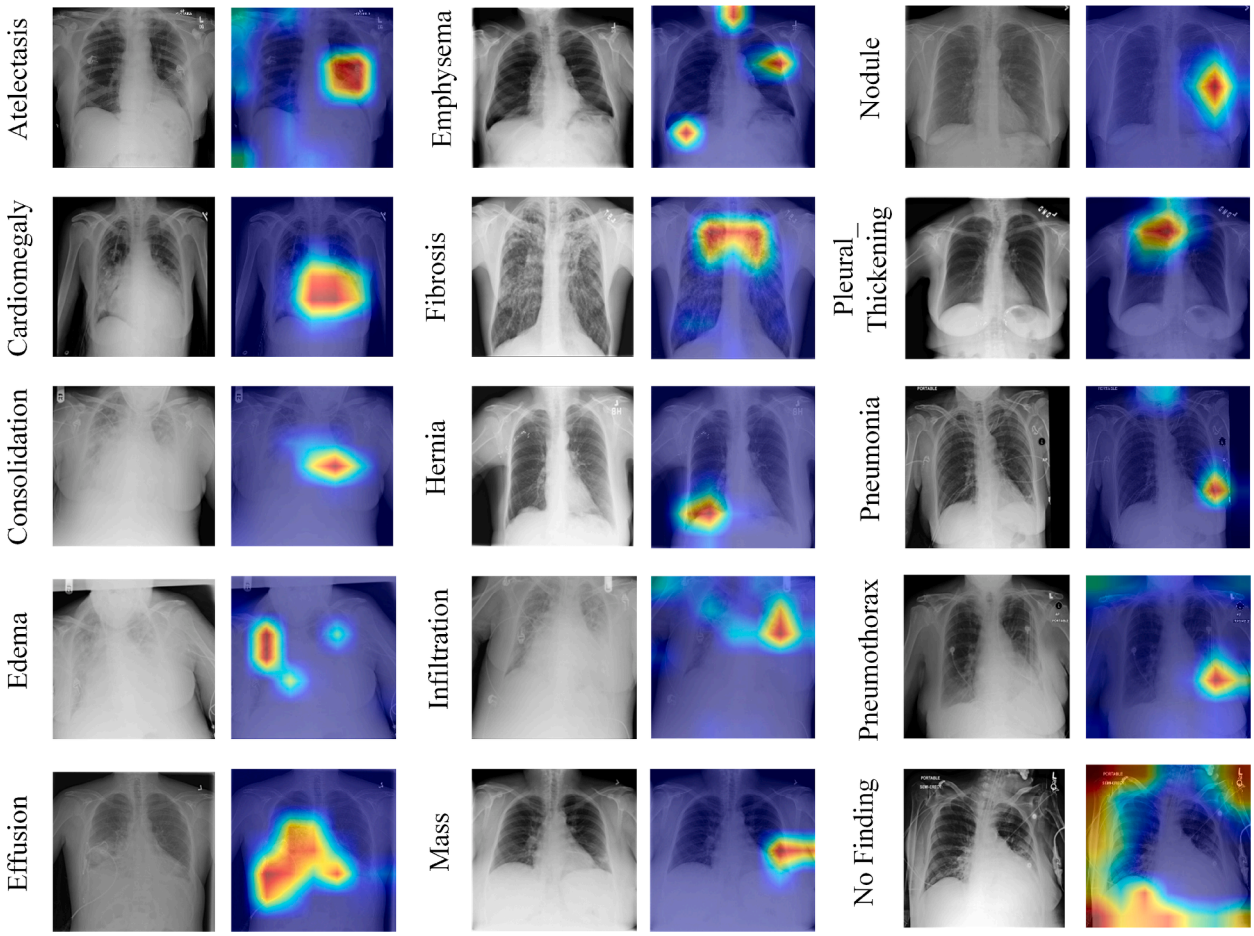


Fig. 9. Attentional heat maps generated by Grad-CAM on our model for selected diseases on the three datasets. The figure shows the visualization results of the attentional map in our model, with each row of three columns from left to right representing the disease name, the original image, and the attentional map of the corresponding disease generated by our model.

supervised learning are utilized to generate more samples of rare diseases to mitigate the effects of data imbalance.

4.6.2 Limitations of labeled co-occurrence matrix

In this paper, we use Graph Convolutional Networks (GCNs) to generate label co-occurrence matrices to capture the relationships between conditions. Although this method can help the model to understand the co-occurrence relationships between labels, this may not be sufficient to comprehensively reflect the complex associations between conditions, especially in terms of differences in conditions and the evolution of the conditions from one patient to another. In addition, the co-occurrence matrix may be too dependent on the distribution of labels in the training set, leading to unstable performance of the model when dealing with new and differently distributed data. More complex relationships between labels can be captured in the future by more advanced relational modeling approaches such as dynamic graph neural

networks or causal inference-based models. Attempts could also be made to introduce more a priori medical knowledge or input from clinical experts in combination with graph convolutional networks to construct more representative models of label relationships.

5. Conclusion

In this paper, we proposed a method that combines anatomical segmentation knowledge and label co-occurrence relationships based on using large kernel convolution to extract image contextual features. Firstly, we pre-trained a UNet from a small dataset with organ-level annotations which benefits from segmenting the regions of chest anatomical in the images. Secondly, we build a Backbone Network by designing a four-stage LKA mechanism and superimpose the former anatomical segmentation results onto the feature maps obtained at each stage to get the fused new feature maps. Thirdly, we consider the graph convolutional network as an independent branch to obtain the co-occurrence matrix among all labels in training datasets. Lastly, we combined the co-occurrence matrix and fused feature maps to consider both label relationships and visual feature information, thereby obtaining the predicted diagnostic results. Experiments on three publicly available large CXR datasets (CheXpert, NIH, and MIMIC-CXR-JPG) and comparisons with existing methods show that our approach achieves better results, demonstrating its effectiveness and generality. There are also some disadvantages, such as our model paying much more attention to the lung and heart regions, which may cause less information about the boundary regions lost. In addition, there are phenomena that long-tail class distribution problems that appeared on the NIH dataset and the MIMIC-CXR-JPG dataset, which makes our model more sensitive to classes with larger sample sizes. How to solve these problems is also a work that needs to be continued to be explored in the future.

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CRediT authorship contribution statement

Xiangxin Zhao: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Xin Wang:** Formal analysis, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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